

MOVING CHARGES AND MAGNETISM

1

Biot-Savart's Law

$$dB \propto \frac{Idl \sin \theta}{r^2}$$

$$dB = \frac{\mu_0}{4\pi} \frac{Idl \sin \theta}{r^2}$$

μ_0 → magnetic permeability of free space or vacuum

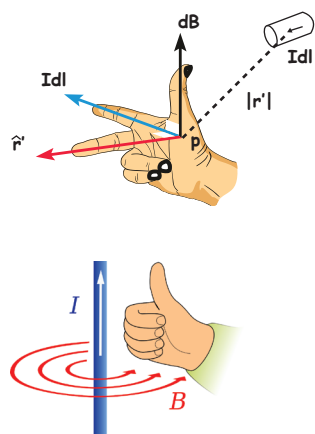
$$\mu_0 = 4\pi \times 10^{-7}$$

In vector form,

$$d\vec{B} = \frac{\mu_0}{4\pi} I \frac{d\vec{l} \times \vec{r}}{r^3}$$

$d\vec{B}$ is perpendicular to both $d\vec{l}$ and $\vec{r}$. BY using right hand screw rule we can find direction of magnetic field

Here, B is into the plane



Magnetic field circulates around the current carrying wire

Formula of Field due to straight wire

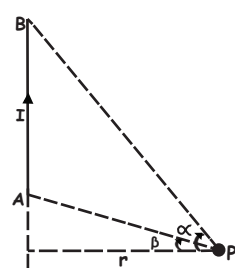
1) At point P

B is into the plane

$$B = \frac{\mu_0 I}{4\pi r} (\sin \alpha + \sin \beta), \otimes$$

2)

Extend AB downwards and draw a perpendicular from P



$$B = \frac{\mu_0 I}{4\pi r} (\sin \alpha - \sin \beta), \otimes$$

3) Wire of infinite length

$$\alpha = 90^\circ \text{ and } \beta = 90^\circ$$

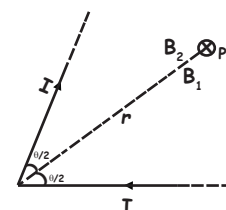
$$B = \frac{\mu_0 I}{4\pi r} (\sin 90^\circ + \sin 90^\circ) = \frac{\mu_0 I}{2\pi r}, \otimes$$

4) Wire of semi infinite length

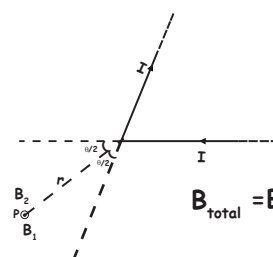
$$B = \frac{\mu_0 I}{4\pi r} (\sin 90^\circ + \sin 0^\circ) = \frac{\mu_0 I}{4\pi r}$$

$$B(P_1) = B(P_2) = 0$$

Field Due to bent wire

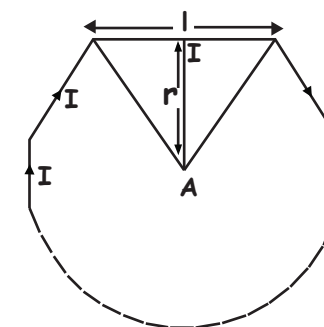


$$B_{\text{total}} = B_1 + B_2 = 2 \times \frac{\mu_0 I}{4\pi r} \frac{[1 + \cos \theta/2]}{\sin \theta/2}, \odot$$



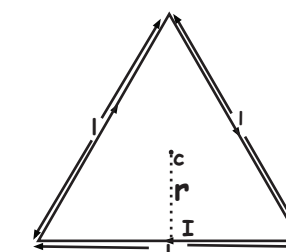
$$B_{\text{total}} = B_1 + B_2 = 2 \times \frac{\mu_0 I}{4\pi r} \frac{[1 - \cos \theta/2]}{\sin \theta/2}, \otimes$$

Field due to polygon



$$B_A = \frac{\mu_0 n I}{\pi l} \tan \frac{\pi}{n} \sin \frac{\pi}{n}$$

eg: equilateral triangle

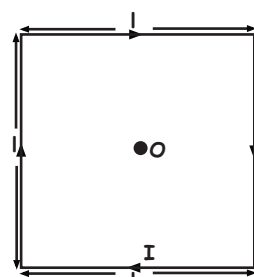


$$B_C = \frac{9}{2} \frac{\mu_0 I}{\pi l}$$

If a total length l is bent as an equilateral triangle, each side has length l/3 then,

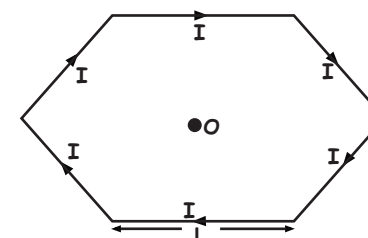
$$B' = \frac{27}{2} \frac{\mu_0 I}{\pi l}$$

Square



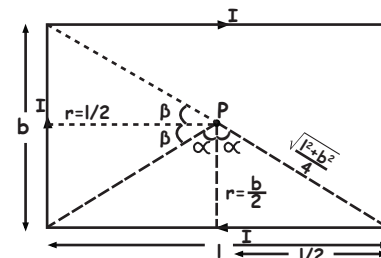
$$B_O = \frac{2\sqrt{2} \mu_0 I}{\pi l}$$

Hexagon



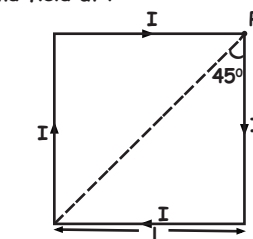
$$B_O = \frac{\sqrt{3} \mu_0 I}{\pi l}$$

Rectangle



$$B_P = \frac{2\mu_0 I}{\pi} \left(\frac{\text{diagonal}}{\text{area}} \right) = 2\mu_0 I \frac{\sqrt{a^2 + b^2}}{ab}$$

Note:- To find field at P



Here P is a point at vertex

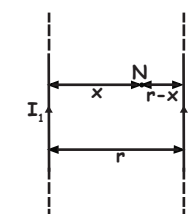
$$B_P = \frac{\mu_0 I}{2\sqrt{2} \pi l}$$

Neutral Points

Points at which magnetic field becomes zero are called neutral points.

Case 1: Parallel wires carrying current in same direction ($I_1 > I_2$)

$$\therefore x = \frac{I_1 r}{I_1 + I_2}$$

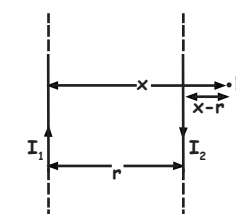


If y is the distance from 2nd conductor then

$$y = \frac{I_2 r}{I_1 + I_2}$$

Case 2: Parallel wires carrying currents in opposite direction ($I_1 > I_2$)

$$x = \frac{I_1 r}{I_1 - I_2}$$

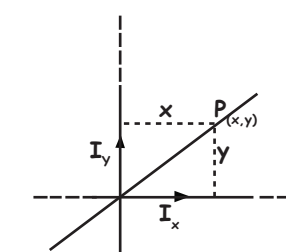


Case 3: Wires perpendicular to each other

Field will be zero on all points of the line OP

Comparing with $y = mx + C$ we get.

$$\text{slope } m = \frac{I_x}{I_y}$$



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Wire perpendicular to plane

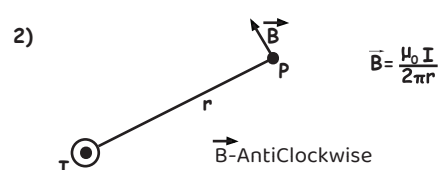
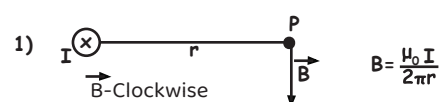
1) Current directed out of plane



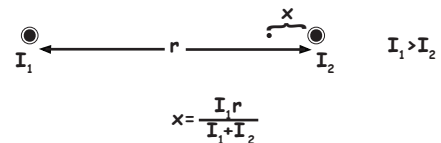
2) Current directed into the plane



Direction of field

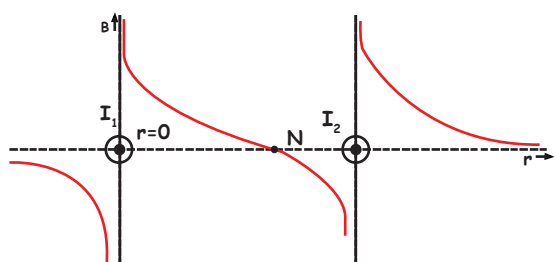


Neutral Point

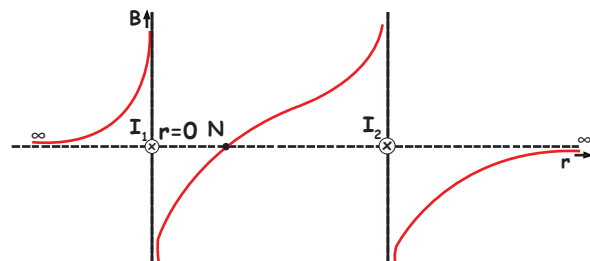


Graphs

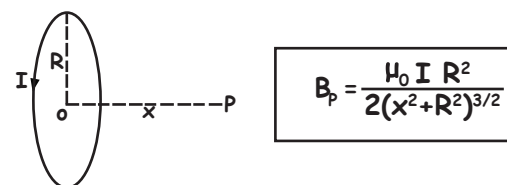
1) I out of the plane, $I_1 > I_2$



2) I into the plane, $I_1 < I_2$



Field due to a circular ring & Field at the axis of a ring



Every current carrying loop acts as a magnetic dipole

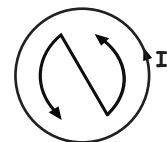
$$\vec{M} = I\vec{A}$$

Direction of A is determined using right hand thumb rule

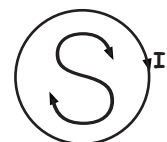
$$\text{Here, } \vec{M} = I\vec{A} \Rightarrow M = I\pi R^2$$

$$B = \frac{\mu_0}{4\pi} \frac{2M}{(x^2 + R^2)^{3/2}}$$

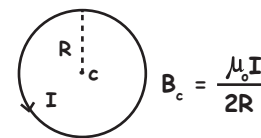
If current is flowing in anticlockwise direction



If current is flowing in clockwise direction



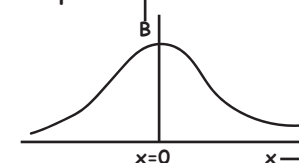
Field at the centre of ring



Field at the axis in terms of field at center

$$B_{\text{axis}} = \frac{B_c}{\left(1 + \frac{x^2}{R^2}\right)^{3/2}}$$

Graph



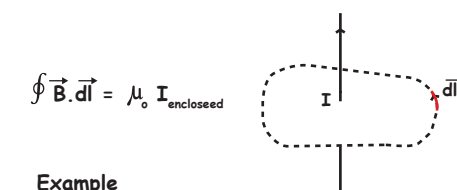
Percentage change in field with respect to centre for the points on the axis

$$\% \text{ change in } B = 1 - \left(1 + \frac{x^2}{R^2}\right)^{-3/2} \times 100\%$$

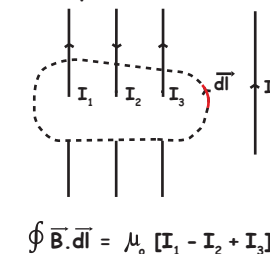
$$\text{For change} < 10\% = \left(\frac{3}{2} \times \frac{x^2}{R^2}\right) \times 100\%$$

Ampere's Circuital Law

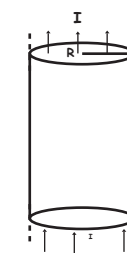
The line integral of magnetic field over a closed loop is equal to μ_0 times the total current enclosed by the loop



Example



Solid Cylinder



Outside : $r > R$

$$B_{\text{out}} = \frac{\mu_0 I}{2\pi r}$$

$$B_{\text{out}} \propto \frac{1}{r}$$

On the surface $r = R$

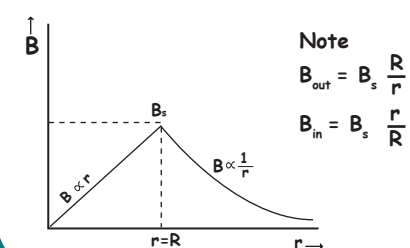
$$B_s = \frac{\mu_0 I}{2\pi R}$$

Inside ($r < R$)

$$B_{\text{in}} = \left(\frac{\mu_0 I}{2\pi R^2}\right) r$$

$$B_{\text{in}} \propto r$$

Graph

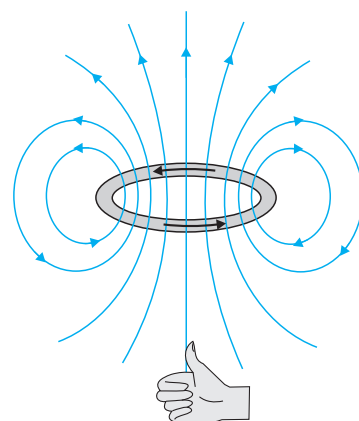


Note

$$B_{\text{out}} = B_s \frac{R}{r}$$

$$B_{\text{in}} = B_s \frac{r}{R}$$

Magnetic field lines for a current loop



$$\text{If there are } N \text{ loops, } B = \frac{\mu_0 N I R^2}{2(x^2 + R^2)^{3/2}}$$



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Field at the Centre due to circular arc

Field at centre of full Circle,

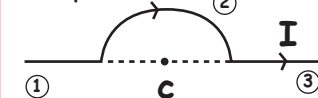
$$B = \frac{\mu_0 I}{2r}$$

Then field at centre of arc,

$$B = \frac{\mu_0 I}{4\pi} \frac{\theta}{r}$$

where, θ is in radian

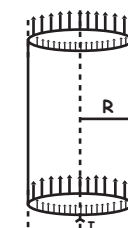
Example



$$B_c = B_1 + B_2 + B_2$$

$$= \frac{\mu_0 I}{4\pi} \frac{\pi}{r} = \frac{\mu_0 I}{4r}$$

Hollow cylinder (pipe)



Outside : $r > R$

$$B_{\text{out}} = \frac{\mu_0 I}{2\pi r}$$

$$B_{\text{out}} \propto \frac{1}{r}$$

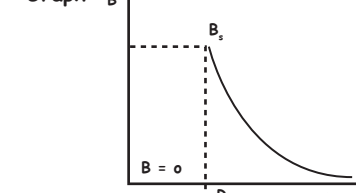
On the surface : $r = R$

$$B_s = \frac{\mu_0 I}{2\pi R}$$

Inside : $r < R$

$$B_{\text{in}} = 0$$

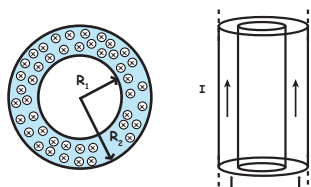
Graph



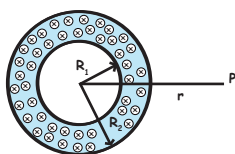
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Annular pipe

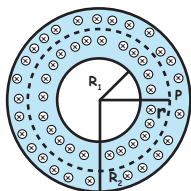


• Outside ($r > R_2$)



$$B_{out} = \frac{\mu_0 I}{2\pi r}$$

• In between $R_1 < r < R_2$



$$I_{enc} = \frac{I}{\pi(R_2^2 - R_1^2)} \times \pi(r^2 - R_1^2)$$

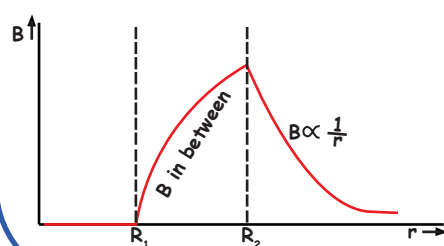
$$B_{btw} = \frac{\mu_0}{2\pi r} \times I \frac{(r^2 - R_1^2)}{(R_2^2 - R_1^2)}$$

Inside

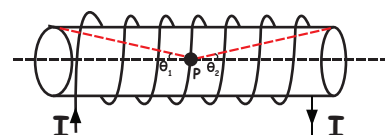
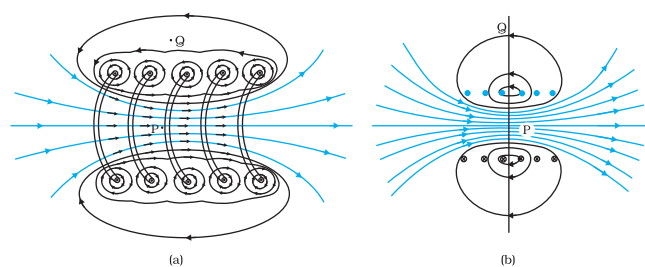
$$r < R_1$$

$$B_{in} = 0$$

Graph



Solenoid



$$B_p = \frac{1}{2} \mu_0 n I (\cos\theta_1 + \cos\theta_2)$$

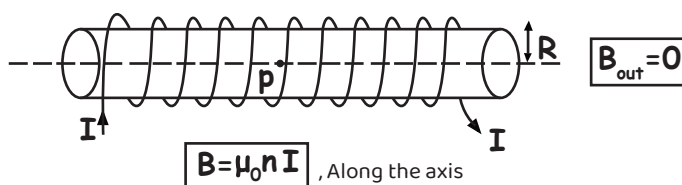
where $n \rightarrow$ no. of turns per unit length

Tightly packed long solenoid

Long solenoid: Radius is small compared to the length of the solenoid

For a long solenoid the middle region will have a uniform magnetic field

1) At centre



2) End point

$$\theta_1 = 0 \quad \theta_2 = 90^\circ$$

$$B_{end} = \frac{1}{2} \mu_0 n I = \frac{B_{axis}}{2}$$

Charged particle in magnetic field

Magnetic Force on particle

$$\vec{F}_m = q(\vec{v} \times \vec{B})$$

Case $\vec{v} \perp \vec{B}$

Path: Circle

Radius of circular path

$$i) F_m = \frac{mv^2}{r}$$

$$\Rightarrow qvB = \frac{mv^2}{r}$$

$$\Rightarrow r = \frac{mv}{qB}$$

ii) Momentum

$$p = mv = \sqrt{2mk}$$

$$\Rightarrow r = \frac{\sqrt{2mk}}{qB}$$

If charged particle at rest is accelerated by a voltage of 'V' volt then,

$$K.E = qV$$

$$r = \frac{\sqrt{2mqV}}{qB}$$

If $K_i \neq 0$, then $K_f = qV + K_i$

$$r = \frac{\sqrt{2mk}}{qB} = \frac{\sqrt{2m(qV + K_i)}}{qB}$$

Time Period

$$T = \frac{2\pi m}{qB}$$

$$\Rightarrow T \propto r^0$$

$$T \propto v^0$$

We have $\frac{q}{m}$ = specific charge

$$\Rightarrow T \propto \frac{1}{\text{specific charge}}$$

Calculation of ratio of radii

1) All particles are projected with same speed:

$V = \text{Constant}$

$$r = \frac{mv}{qB} \Rightarrow r \propto \frac{m}{q}$$

$$r_p : r_d : r_\alpha = \frac{1}{e} : \frac{2}{e} : \frac{4}{2e} = 1:2:2$$

2) All particles are projected with same momentum

$$r = \frac{p}{qB} \Rightarrow r \propto \frac{1}{q}$$

$$r_p : r_d : r_\alpha = \frac{1}{e} : \frac{1}{e} : \frac{1}{2e} = 2:2:1$$

As radius $\uparrow$ Curvature $\downarrow$

3) All particles are projected with same kinetic energy

$$r = \frac{mv}{qB} = \frac{\sqrt{2mk}}{qB}$$

$$\Rightarrow r \propto \frac{\sqrt{m}}{q}$$

$$r_p : r_d : r_\alpha = \frac{\sqrt{1}}{e} : \frac{\sqrt{2}}{e} : \frac{\sqrt{4}}{2e} = 1 : \sqrt{2} : 1$$

4) All particles are projected by same accelerating potential

$$r = \frac{\sqrt{2mqV}}{qB} \Rightarrow r \propto \sqrt{\frac{m}{q}}$$

$$r_p : r_d : r_\alpha = \sqrt{\frac{1}{e}} : \sqrt{\frac{2}{e}} : \sqrt{\frac{4}{2e}} = 1 : \sqrt{2} : \sqrt{2}$$

Helical path & pitch

$\vec{v}$ makes angle θ with $\vec{B}$ ($\theta \neq 0, \pi, \pi/2$)

Path of charge is Helical

$$\vec{v} \begin{cases} V_{\parallel} \rightarrow \parallel \text{ to } \vec{B} \\ V_{\perp} \rightarrow \perp \text{ to } \vec{B} \end{cases}$$

1) Radius

$$R = \frac{mv_{\perp}}{qB}$$

$$R = \frac{mv \sin \theta}{qB} = \frac{\sqrt{2mqV}}{qB} \sin \theta$$

2) Time period

$$T = \frac{2\pi m}{qB}$$

$$\begin{aligned} \text{3) Pitch} &= V_{\parallel} \times T \\ &= V \cos \theta \times \frac{2\pi m}{qB} \\ &= 2\pi \left(\frac{mv}{qB} \right) \cos \theta \\ &= 2\pi \frac{\sqrt{2mk}}{qB} \cos \theta \\ &= 2\pi \frac{\sqrt{2mqV}}{qB} \cos \theta \end{aligned}$$

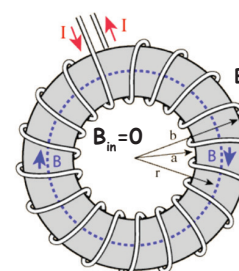


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Toroid

r is the mean radius



$$r = \frac{a_1 + b_2}{2}$$

$$B_{out} = 0$$

$$B_{inside} = 0$$

$$B_{in \text{ between}} = \mu_0 n I$$

unit and dimension of B

$$F = qvB \sin \theta$$

$$B = \frac{F}{qv \sin \theta}$$

$$[B] = \frac{MLT^{-2}}{AT \times LT^{-1}} = MA^{-1}T^{-2}$$

$$\text{Unit} \rightarrow kgA^{-1}s^{-2} = \text{Tesla (T)}$$

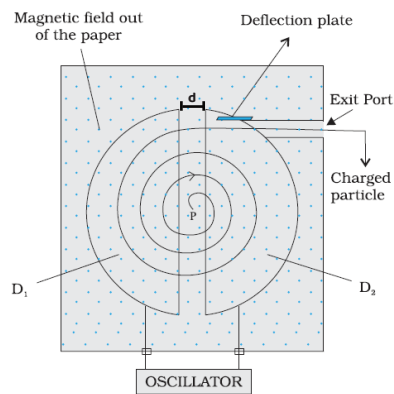
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Cyclotron



1) Maximum kinetic energy

$$K_{\max} = \frac{q^2 B^2 R_{\text{dee}}^2}{2m}$$

2) Number of oscillations, N

$$\text{Work done} = \Delta K$$

$$N \times qEd = \frac{q^2 B^2 R^2}{2m}$$

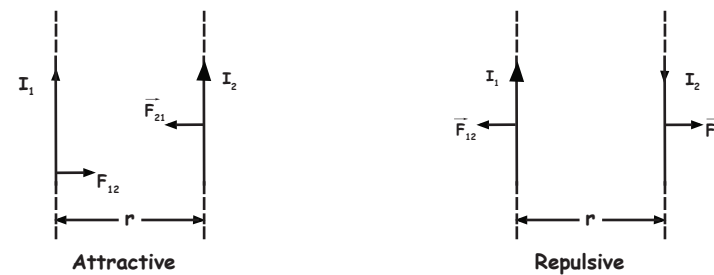
E is in the plane of the dees

$$3) \text{ Time period of revolution} = \frac{2\pi m}{qB}$$

$$4) \text{ Cyclotron frequency } \nu_c = \frac{qB}{2\pi m}$$

Force between parallel conductors

1) Force between two long parallel current conductors

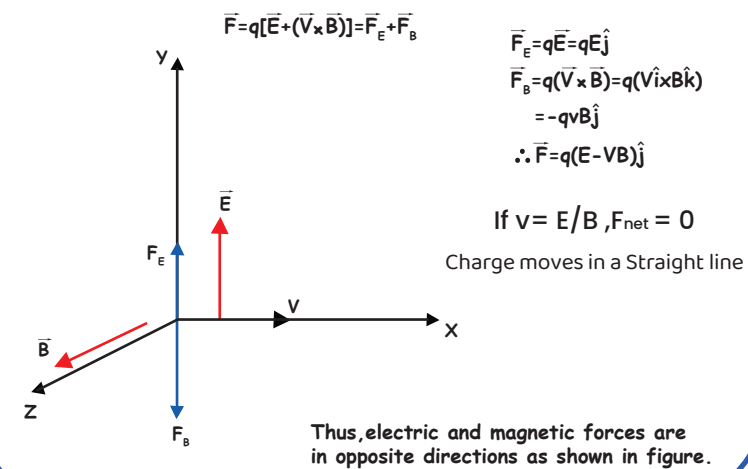


Parallel currents attract each other. Anti parallel currents repel each other.
Force per unit length of conductor

$$F_{12} = F_{21} = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{r}$$

$$\text{Net force on 'l' length of conductor} = \frac{\mu_0}{4\pi} \frac{2I_1 I_2}{r} \times l$$

Motion of charged particle in Crossed Electric and Magnetic field $(\vec{E} \perp \vec{B})$ [velocity selector]



Force on current carrying Conductor in magnetic field

consider a small element $d\vec{l}$ of the conductor

$$d\vec{F} = I(d\vec{l} \times \vec{B})$$

To find resulting force

$$\int d\vec{F} = \int I(d\vec{l} \times \vec{B})$$

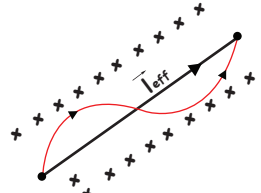
$$= I \int d\vec{l} \times \vec{B}$$

In uniform field

$$\vec{F} = I \left(\int d\vec{l} \right) \times \vec{B} = I(\vec{l} \times \vec{B})$$

The above equation can be used in the following situations

1)



$$\vec{F} = I(\vec{l}_{\text{eff}} \times \vec{B}) = I l_{\text{eff}} B \sin \theta$$

2) $\vec{l}_{\text{eff}}$ perpendicular to $\vec{B}$

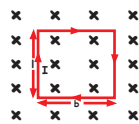
$$\theta = 90^\circ$$

$$F_{\max} = I l_{\text{eff}} B$$

For closed Loop in uniform field

$$\text{Here, } l_{\text{eff}} = 0$$

$$\therefore \vec{F} = 0$$



Note:

1) The constant, $\frac{K}{NAB}$ is called galvanometer constant or current reduction factor of the galvanometer.

2) Figure of merit of a galvanometer

$$G = \frac{I}{\Phi} = \frac{K}{NAB}$$

Lorentz force

A charge q in an electric field $\vec{E}$ experiences the electric force

$$\vec{F} = q\vec{E}$$

The magnetic force experienced by the charge q moving with velocity $\vec{V}$ in the magnetic field $\vec{B}$ is given by

$$\vec{F}_m = q(\vec{V} \times \vec{B})$$

The total force, or the Lorentz force, experienced by the charge q due to both electric and magnetic field is given by

$$\vec{F} = q[\vec{E} + (\vec{V} \times \vec{B})]$$

Moving Coil Galvanometer (MCG)

$$\tau = NIAB \sin \theta$$

Sensitivity of a Galvanometer

$$\text{Current sensitivity, } I_s = \frac{\Phi}{I} = \frac{NAB}{K}$$

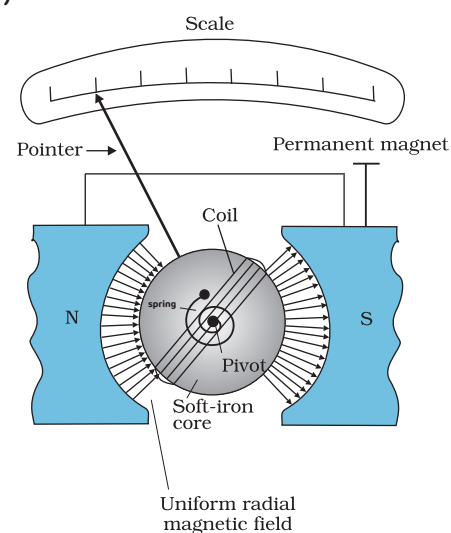
Where $K \rightarrow$ torsional constant

$\Phi \rightarrow$ deflection of galvanometer

Voltage Sensitivity

$$\text{Voltage Sensitivity, } V_s = \frac{\Phi}{V} = \frac{\Phi}{IR} = \frac{NAB}{KR}$$

$$\text{Voltage Sensitivity} = \frac{\text{Current Sensitivity}}{R}$$



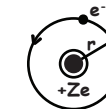
Magnetic dipole moment of a revolving electron

$$I = \frac{e}{T}$$

T is the time period of revolution

$$T = \frac{2\pi r}{v}$$

$$I = \frac{ev}{2\pi r}$$



There will be a magnetic moment, usually denoted by μ associated with this circulating current

$$\mu = I \pi r^2 = \frac{eVr}{2}$$

$$\mu = \frac{e}{2m_e} (m_e v r) = \frac{e}{2m_e} L$$

$$\vec{L} = \vec{r} \times \vec{p} = r m v \sin \theta = m v r$$

In vector form

$$\vec{\mu} = \frac{-e}{2m_e} \vec{L}$$

The negative sign indicates that the angular momentum of the electron is opposite in direction to the magnetic moment

The ratio of magnetic moment to the angular momentum is called gyromagnetic ratio

$$\frac{\mu}{L} = \frac{e}{2m_e}$$

Its value is a constant and is equal to $8.8 \times 10^{-10} \text{ C/kg}$ for an electron

According to Bohr's quantization condition, angular momentum assumes a discrete set of values, namely,

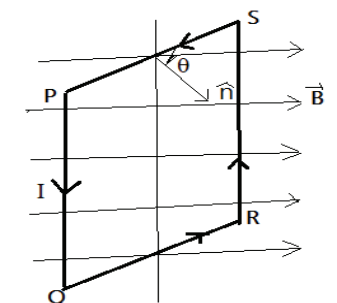
$$L = \frac{nh}{2\pi} \text{ where } n = 1, 2, 3, \dots$$

$h \rightarrow$ planck's constant

Taking $n=1$, we get,

$$\mu_{\min} = \frac{e}{4\pi m_e} h = 9.27 \times 10^{-24} \text{ Am}^2$$

Torque on a current loop in a uniform magnetic field



$$\text{Then } \tau = NIAB \sin \theta$$

Special cases:

i) When $\theta = 0^\circ$

$\tau = 0$, i.e., the torque is minimum when the plane of the loop is perpendicular to the magnetic field

ii) When $\theta = 90^\circ$

$\tau = NIAB$ i.e., the torque is maximum when the plane of the loop is parallel to the magnetic field. Thus,

$$\tau_{\max} = NIAB$$